Motorcycle Shop
E-Commerce System

Name:
Benitez, Alwin C.
Roduta, Kelly Laurence Q.

Section:
BSIT – S – 3A

Submitted to:
Mr. Dalisay

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FUNCTIONAL REQUIREMENTS

The system shall undergo CRUD (Create, Read, Update, Delete) operations and provide e-commerce functionalities for a motorcycle shop, including user management, product management, shopping cart, order processing, installment payments, and inventory restocking with expense tracking. The system is developed using the MERN stack (MongoDB, Express.js, React, and Node.js) and supports three roles: Admin, Inventory Manager, and Customer, ensuring secure access and proper task distribution.

1. Functional Requirements

1.1 User Account Management

This module handles user registration, secure login, and account maintenance. It ensures users can safely access the system and manage their profiles, while administrators retain control over user permissions.

ID	Requirement	Description
FR1.1.1	**User Registration**	The system shall allow users to register an account by providing name, email, password, and address.
FR1.1.2	**Login Validation**	The system shall validate login credentials before granting access.
FR1.1.3	**Profile Update**	The system shall allow users to update personal information such as name, contact number, address, and profile picture. Customers may also upload a valid ID and proof of income, which are used for installment applications.
FR1.1.4	**Password Management**	The system shall allow users to change their password and to reset a forgotten password through a link sent to their email.
FR1.1.5	**Account Deactivation**	The system shall allow users to deactivate their account.
FR1.1.6	**Admin User Control**	The system shall allow the admin to view, update, or delete any user account.
FR1.1.7	**Role Assignment**	The system shall assign roles during account creation (Admin, Inventory Manager, Customer).

1.2 Product Management

This module enables the administrator to manage motorcycles, motorcycle parts, accessories, and riding gear available for purchase. It ensures that product information remains updated and accurately displayed to customers.

ID	Requirement	Description
FR1.2.1	**Add Product**	The system shall allow admin to add new products including name, description, category, brand, compatible motorcycle models, stock, price, image, and, optionally, the installment plans the product is eligible for.

ID	Requirement	Description
FR1.2.2	**Display Products**	The system shall display a list of all available products to customers.
FR1.2.3	**Update Product**	The system shall allow admin to update product details, including its installment eligibility and the minimum price required to qualify for installment.
FR1.2.4	**Archive/Delete Product**	The system shall allow admin to archive or delete products.
FR1.2.5	**Product Search**	The system shall allow product searching by name, brand, category, or compatible motorcycle model.
FR1.2.6	**Product Images**	The system shall allow admin to upload multiple images per product.
FR1.2.7	**Product Details**	The system shall display a product details page showing the product's description, price, stock availability, images, and reviews.
FR1.2.8	**Price Range Filter**	The system shall allow customers to filter products by price range.
FR1.2.9	**Pagination**	The system shall display the product list in pages.
FR1.2.10	**Admin Product List**	The system shall provide the admin with a product management page listing all products, from which the admin can update or delete each product.

1.3 Category and Brand Management

This module organizes products by categories (e.g., Motorcycles, Engine Parts, Brakes, Tires and Wheels, Helmets, Riding Gear) and brands to enhance navigation and browsing experience.

ID	Requirement	Description
FR1.3.1	**Manage Categories**	The system shall allow admin to create, update, and delete product categories (e.g., Motorcycles, Engine Parts, Brakes, Tires and Wheels, Helmets, Riding Gear, Accessories, Lubricants).
FR1.3.2	**Assign Category**	The system shall allow each product to be assigned to a category.
FR1.3.3	**Manage Brands**	The system shall store brands (e.g., Honda, Yamaha, Suzuki, Kawasaki, CFMOTO) in the database.
FR1.3.4	**Filter by Brand**	The system shall allow filtering of products by brand.
FR1.3.5	**Filter by Category**	The system shall allow filtering of products by category.

1.4 Shopping Cart Management

This module allows customers to temporarily store selected products before completing a purchase. It improves shopping convenience and helps track order totals.

ID	Requirement	Description
FR1.4.1	**Add to Cart**	The system shall allow customers to add products to their shopping cart.
FR1.4.2	**Update Quantity**	The system shall allow customers to update item quantity in the cart.
FR1.4.3	**Remove from Cart**	The system shall allow customers to remove products from the cart.
FR1.4.4	**Cart Summary**	The system shall display the cart total price including applicable taxes and, for installment-eligible items, the starting monthly payment.
FR1.4.5	**Save Cart**	The system shall persist the cart for logged-in users across sessions.

1.5 Order and Checkout Management

This module processes customer purchases and tracks order status. It ensures secure and organized order handling from placement to delivery, including orders paid through installment.

ID	Requirement	Description
FR1.5.1	**Place Order**	The system shall allow customers to place orders from their shopping cart.
FR1.5.2	**Shipping Information**	The system shall require customers to provide shipping address, city, phone number, postal code, and country before placing an order.
FR1.5.3	**Order Confirmation**	The system shall display an order summary showing the items, items price, shipping price, tax, and total price for confirmation before payment. For installment orders, it shall also show the plan, downpayment, financed amount, interest and fees, monthly amount, term, and total installment cost.
FR1.5.4	**Record Order Details**	The system shall record order details such as items, total price, date, and payment method. For installment orders, it shall also record the plan, downpayment, and payment schedule.
FR1.5.5	**Payment Methods**	The system shall allow cash on delivery, online payment and installment as payment methods.
FR1.5.6	**Order Status**	The system shall assign an order status: Pending, Processing, Shipped, Delivered, or Cancelled. Installment orders shall also use For Approval and Awaiting Downpayment, and shall not move to Processing or be released for delivery until the plan is approved and the downpayment is confirmed.
FR1.5.7	**Order History**	The system shall allow customers to view their order history and the details of each order. For installment orders, customers shall also see their payment schedule, remaining balance, and payment history.
FR1.5.8	**Admin Order Management**	The system shall allow admin to view all orders, update order status, and delete orders.

ID	Requirement	Description
FR1.5.9	**Order Cancellation**	The system shall allow customers to cancel orders with Pending status. For installment orders that already have a downpayment, the system shall record any refund or forfeited downpayment according to the store's policy.
FR1.5.10	**Mailing / Messaging**	The system shall notify or send mail through Mailtrap to the user regarding their order summary as well as when the order is updated to a specific status (e.g., delivered). For installment orders, it shall also send mail on approval or rejection, downpayment confirmation, upcoming or missed due dates, and payment receipts.
FR1.5.11	**Installment Plan Selection**	The system shall allow customers paying by installment to choose an eligible plan and a downpayment within the allowed limits at checkout.

1.6 Inventory and Restocking Management

This module ensures accurate stock levels and records restocking transactions to prevent product shortages. It also tracks expenses related to product purchases from suppliers.

ID	Requirement	Description
FR1.6.1	**Record Restocking**	The system shall allow the inventory manager to record restocking of products.
FR1.6.2	**Supplier Details**	The system shall store supplier details including name, contact information, and supplied product lines.
FR1.6.3	**Expense Tracking**	The system shall track restocking cost per product and compute total expenses.
FR1.6.4	**Update Stock**	The system shall update stock levels after restocking or after a sale. For installment orders, stock shall be reserved once the installment is approved.
FR1.6.5	**Inventory History**	The system shall record a full inventory transaction history.
FR1.6.6	**Low Stock Alert**	The system shall notify the inventory manager when a product's stock falls below a defined threshold.

1.7 Review and Rating System

This module enhances product credibility by allowing customers who have purchased a product to provide feedback and share their experiences.

ID	Requirement	Description
FR1.7.1	**Star Rating**	The system shall allow customers to rate products on a 1–5 star scale.
FR1.7.2	**Write Review**	The system shall allow customers to write product reviews including title and body text.

ID	Requirement	Description
FR1.7.3	**Display Reviews**	The system shall display customer reviews and average rating per product.
FR1.7.4	**Verified Purchase Badge**	The system shall mark reviews from verified purchasers. For installment orders, the badge applies once the order is Delivered.
FR1.7.5	**Admin Moderation**	The system shall allow admin to view, hide, or delete inappropriate reviews.
FR1.7.6	**Edit Own Review**	The system shall allow a customer who reviews a product again to update their existing review instead of adding a second one.

1.8 Role and Access Control

This module restricts access to system functionalities based on predefined user roles, ensuring data privacy and security.

ID	Requirement	Description
FR1.8.1	**Role Support**	The system shall support three roles: Admin, Inventory Manager, and Customer.
FR1.8.2	**Feature Restriction**	The system shall restrict access to features based on user role. Installment approval and manual payment recording shall be restricted to the Admin.
FR1.8.3	**Secure Login/Logout**	The system shall require secure login and logout using token-based authentication.
FR1.8.4	**Unauthorized Access Prevention**	The system shall prevent unauthorized module access.
FR1.8.5	**Payment Data Security**	The system shall not store card details.

1.9 Reporting

This module provides summarized data for business decision-making and system monitoring. Hence, charts or graphs will be utilized for this.

ID	Requirement	Description
FR1.9.1	**Sales Report**	The system shall generate a sales report by date range.
FR1.9.2	**Inventory Report**	The system shall generate an inventory stock report.
FR1.9.3	**Expense Report**	The system shall generate an expense report for restocking.
FR1.9.4	**Order Summary**	The system shall generate a customer order summary report.
FR1.9.5	**Top Products Report**	The system shall generate a report of best-selling products by category.

ID	Requirement	Description
FR1.9.6	**Admin Dashboard**	The system shall display summarized sales, order, and product data to the admin on a dashboard using charts, including installment collections and overdue or defaulted accounts.
FR1.9.7	**Installment Reports**	The system shall generate reports on payment collections, outstanding balances (receivables), and overdue accounts.

1.10 Installment Management

This module covers the installment option for customers who prefer to pay over time instead of in full. Plans such as 12 months with a fixed monthly payment, or 36 months with a lower downpayment, are examples that the admin can configure. This module covers the plans, approval, schedules, and collections behind them.

ID	Requirement	Description
FR1.10.1	**Manage Installment Plans**	The system shall allow the admin to create, update, and archive installment plans with a term (e.g., 3, 6, 12, 24, or 36 months), interest rate, minimum downpayment percentage, and optional processing fee.
FR1.10.2	**Amortization Computation**	The system shall compute the financed amount, interest, fees, monthly payment, and total cost for a chosen plan and downpayment, and show the breakdown before the customer confirms.
FR1.10.3	**Application and Approval**	The system shall require customers who choose installment to submit supporting details (e.g., valid ID, proof of income) for admin review, and shall allow the admin to approve or reject the application and record the reason for rejection.
FR1.10.4	**Payment Schedule**	The system shall generate a payment schedule listing each due date, amount, and payment status.
FR1.10.5	**Record Payments**	The system shall record each payment (amount, date, method, reference number), update the remaining balance, and generate a receipt that can be viewed and emailed. The admin may also record payments made manually (e.g., cash or bank deposit).
FR1.10.6	**Payment Status**	The system shall track each scheduled payment as Unpaid, Paid, or Overdue, and each installment order as Unpaid, Partially Paid, Paid, or Defaulted.
FR1.10.7	**Late Payment Penalty**	The system shall, if enabled by the admin, apply a configurable penalty (fixed amount or percentage) to overdue payments after an optional grace period.
FR1.10.8	**Early Payoff**	The system shall allow customers to pay ahead or settle the remaining balance early.
FR1.10.9	**Default Handling**	The system shall allow the admin to mark an installment account as Defaulted after a set number of missed payments and shall flag it on the dashboard.